

Simulating the Motion of Trapped Anti-Hydrogen

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Abstract

In this report we explore different methods for simulating the motion of an ensemble of particle-like objects. This includes the numerical integration of analytically unsolvable equations, symplectic leapfrog integration and random variable distribution subject to a cumulative distribution function. We run a simulation for the non-adiabatic heating of particles via Gaussian beam "kicking" and fit the resulting data to exponential models, returning a "mean escape time" of 7740 ± 200 s, an upper temperature limit of 10.51 ± 0.18 ms⁻¹ and inferring properties of the simulation.

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1 Background and Motivation

1.1 A Brief History on the Discovery of Antimatter

The existence of anti-particles in modern physics was first postulated in 1928 by Paul Dirac who, while postulating the relativistic Schrödinger equation for a quantum mechanical system with an intrinsic spin, predicted that the wave-function solutions for a point like electron allowed for a positive e charge as well as the expected $-e$ solution, as well as a negative solution to the quantum wave function operator $\hat{W}$,

$$\hat{W} = i\hbar \frac{\delta}{\delta t}, \quad (1)$$

where all symbols have their usual meanings [1]. Although initially sidelined by Dirac, this implication was later developed by J. Robert Oppenheimer, who argued for the existence of an elementary particle with the same mass as an electron but with opposite charge [2] (against Dirac's suggestion that the positive charge solutions corresponded to protons). Dirac then, in 1931, mathematically predicted the existence of anti-electrons (as Oppenheimer had argued), pair production of electron-positron pairs, as well as their mutual annihilation upon interaction [3]. The positron was then observed for the first time in 1932 when Carl Anderson first photographed the track of a positively charged particle, with a mass in the same order of magnitude for an electron, when measuring high energy cosmic rays using a cloud chamber [4]. It is now a well known implication of quantum field theory that every elementary particle has its own corresponding anti-particle, with some particles such as the photon being their own antiparticle [5].

1.2 The Matter-Anti-Matter Symmetry Problem

Our standard particle model of physics requires that all physical processes are invariant under independent inversions of charge, parity or the reversal of time. If this rule were to be

strictly true, there would be equal amounts of matter and anti-matter present in the universe. This, however, seems not to be the case, and we instead appear to have a small surplus in matter to about one part in a billion, leaving (post annihilation) the entire universe as we know it [6]. We cannot directly observe if the universe is matter dominated; from quantum field theory we know that all physical systems are invariant under a combined Charge, Parity and Time inversion (henceforth C P and T) [7], meaning any measurements (for example, spectroscopic) that would indicate an astrophysical body were composed of anti-matter or matter, would be inconclusive. However, a key piece of evidence for concluding that the universe is in fact matter-dominated is that we do not see large scale annihilation events that would otherwise be common on a cosmological scale. From this we can postulate two possible scenarios for our universe; either all structures composed of matter or anti-matter exist only in separate, distinct regions with no points of contact, or there simply exists a deficit of anti-matter. These scenarios suggest two fundamental symmetry breaks within the Standard Model (henceforth SM):

1. Physical mechanisms favor the formation of matter over anti-matter.
2. Gravity interacts differently with matter and antimatter.

1.3 The Weak Equivalence Principle

For the first case, in 1967 Andrei Sakharov proposed three conditions which must be met for a universe to generate baryons (known as baryogenesis) consistent with the observed imbalance between matter and antimatter. The three conditions are baryon number (B) violation, C and CP symmetry violations and to have interactions out of thermal equilibrium [8]. P and CP violations for leptons and mesons were first presented experimentally in the Wu (1957) [9] and in the Cronin-Fitch (1967) [10] experiments respectively, but CP violation was only recently observed for baryon decay [11]. Moreover, baryogenesis may still

result in a discrepancy in net baryon production when incorporated into the SM that cannot currently be explained.

The latter case would require a violation of the Einstein weak equivalence principle (henceforth EWEP), which dictates that an object’s inertial and gravitational masses are equal, and that the outcome of any (non-gravitational) experiment is independent of its velocity relative to a gravitational field, as well as when and where it is performed in the field. Consequently, an anti-particle will behave identically in a matter gravitational field to its particle counterpart; as opposed to CPT symmetry asserting only that an antiparticle will behave in an anti-matter gravitational field as a particle would in a matter field, but nothing of an anti-particle’s behavior in a matter field or vis-a-versa [12]. The violation of the EWEP and the resulting new theory of gravity may reproduce the matter-anti-matter asymmetry needed for baryogenesis [13]. In addition, the EWEP, as a product of general relativity (henceforth GR), has yet to be unified with the SM, making it an attractive potential solution to the model’s shortcomings and motivates our use of anti-matter as a probe into the predictions of GR and the SM.

2 Anti-Hydrogen as an EWEP Probe

The EWEP has been successfully tested experimentally for matter to a sensitivity in the order of 10^{-15} [14], with further proposed experiments (e.g. STEP, [15]) designed to measure EWEP violation to such a precision that a null result would be just as profound as a positive one [16]; suggesting gravity is fundamentally different from the other forces in the SM. Performing these same measurements on anti-matter has historically been challenging due to the lack of a stable, neutral anti-particle [17] but, with the production and confinement of anti-matter hydrogen atoms in the ALPHA-2 apparatus at CERN now possible [18], these measurements can be extended to test the EWEP for anti-matter. The anti-hydrogen is the simplest system of

atomic anti-matter, and has been studied extensively recent years; from techniques to produce and confine anti-hydrogen atoms [19] to precision measurements of its properties and comparisons with the well-explored hydrogen atom [20, 21]. Continuing with these successes, anti-hydrogen is a forefront candidate for the dynamic gravitational experiments [22] which concern this project.

2.1 Production of Anti-Protons and Positrons for ALPHA

The ALPHA (Anti-Hydrogen Laser Physics Apparatus) is located adjacent to the CERN Anti-Proton Decelerator (AD) complex at CERN, which provides a beam of anti-protons cooled to a kinetic energy of 100 KeV (equivalent to about 1.5% the speed of light) [23]. Anti-proton production begins with an accelerated proton beam (kinetic energies over 20 GeV) incident upon a proton rich iridium target, producing anti-protons among other secondary particles [24] via the reaction,

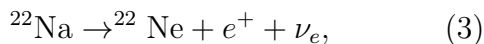
$$p(\text{target}) + p(\text{beam}) \rightarrow p + p + p + \bar{p}, \quad (2)$$

where $p, \bar{p}$ represent protons and antiprotons respectively. The resulting anti-protons are directed towards the AD, where they are then decelerated and collected in four steps via stochastic and then electron cooling down to a kinetic energy of 5.3MeV [25, 26]. The anti-protons are then further decelerated for increased trapping efficiency in a smaller cyclotron, the Extra Low ENergy Antiproton (ELENA) ring [27], producing dense bunches of 100 KeV anti-protons about every 100 seconds for use in all experiments present at CERN simultaneously, 24 hours a day. Experiments performed in the ALPHA apparatus regularly exceed the pulse duration of the AD, as such ALPHA incorporates a standalone anti-proton catching trap, giving the apparatus its own reservoir of accumulated anti-protons independent of the regular AD slowing cycles [28].

Delivered anti-protons are then captured and prepared for anti-hydrogen formation in the ALPHA Catching Trap (CT). For

capture, the CT utilizes a Penning-Malberg trap, which uses strong magnetic ($\mathbf{B}$) and electrostatic ($\mathbf{E}$) fields for radial and axial confinement respectively of charged particles. The longitudinal $\mathbf{B}$ field is generated via a trap-length solenoid, while a series of hollow cylindrical high voltage electrodes are used to generate a strong $\mathbf{E}$ field [29]. The exact electrode stack geometry can be varied to the needs of the trap, with static or time-varying voltages allowing the shape of the potential to be altered to suit the need of the experiment [27]. Anti-protons arrive at the CT with an energy of 100KeV, and must be slowed down to trappable energies; to achieve this, the CT utilizes a series of degrading foils, allowing inelastic collisions between the anti-protons and the foil to slow down the anti-protons to about 5KeV, and increase in energy spread [27]. To confine neutral anti-hydrogen, ALPHA-2 utilizes magnetic confinement; for ready formation of anti-hydrogen, anti-protons must have an energy comparable to the depth of the magnetic well (about 0.5K, equivalent to $50\mu\text{eV}$), slowed down from the (relatively) high delivered energy of about 5KeV. As such, the second job of the CT is to convert the now captured anti-protons into a cold, dense plasma through methods such as electron cooling [30].

Positrons are provided in a standalone instrument at ALPHA, the Positron Accumulator (PA). A man-made source of 22-Sodium (^{22}Na) is used, half life $\tau_{1/2} = 2.6$ years, to produce a uniform delivery of positrons over an experiment duration (about 8-6 hours) [31, 32]. Positrons are emitted via β^+ following the equation,

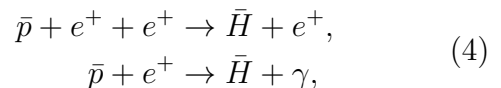


where a positron (e^+) and an electron neutrino (ν_e) are produced. Produced positrons are high in energy and energy range, and must be slowed down and confined before combination with anti-protons; The first step of this process is to cryogenically cool the Na to 8K and moderate the emitted e^+ with a solid neon moderator. The slowed e^+ are then directed into the PA, utilizing a 3-stage Penning-Malberg trap

and molecular nitrogen buffer gas, while varying electrostatic potential to further slow and trap the e^+ . Molecular nitrogen is used due to a high interaction cross-section near the limit for e^+ energy loss, which competes with the main loss mechanism in the PA, the formation of positronium (e^-e^+) [28]. The result is an accumulation of cooled positrons (bunches about every 200s) ready for combination in the Mixing Trap of ALPHA-2.

2.2 Production of Anti-Hydrogen in ALPHA-2

Anti-hydrogen is formed and held in the Mixing and Atom Traps respectively; while they occupy the same space, they are two distinct systems. An overview schematic for the layout of the ALPHA-2 apparatus can be seen in Figure 1. The Mixing Trap uses a Penning-Malmberg trap to manipulate and gently mix the prepared positrons and anti-protons under controlled conditions. The Mixing Penning-Malmberg Trap consists of 3 wells, a central region for mixing and two side regions for particle preparation. Anti-protons are cooled and compressed through coulomb collisions with an adjacent electron plasma (electron cooling), before the electrons are ejected. Positrons are then directed into the other side of the trap and post processed (e.g. further cooled), and the difference in well depth of the two regions slowly reduced until the two plasmas come into contact [28]; during this, higher energy positrons can escape the well, causing further evaporative cooling. A description of the axial potential profile can be seen in Figure 2. The anti-protons and positrons then thermally equilibrate, the anti-protons with the positrons and the positrons with the background radiation through coulomb interactions, and combine in a three-body recombination process [31] or radiative recombination [33]. These processes follow the equations,



respectively, where the γ and e^+ on the RHS carries energy lost from the positron and anti-proton in their recombination. The

cross-section for recombination is proportional to the relative velocity between the constituent particles, and most of the kinetic energy of the formed anti-hydrogen comes from the anti-proton (by mass). The efficiency of trapping the resulting anti-hydrogen and stopping annihilation on the trap walls is determined by its kinetic energy and its formed spin state (which contributes to its magnetic moment). After mixing, any remaining positrons and anti-protons are ejected through opposite sides of the trap.

The Atom Trap cohabits with the Mixing Trap in the ALPHA-2 apparatus, and is responsible for confining the produced neutral anti-hydrogen through the interaction with a magnetic field and the atoms magnetic dipole moment described by,

$$U = -\vec{\mu} \cdot \mathbf{B}, \quad (5)$$

to prevent annihilation with the apparatus walls; where $\vec{\mu}$ and $\mathbf{B}$ are the vector magnetic dipole moment and trap magnetic field respectively, and U is the interaction's potential energy. The Atom Trap is engineered to produce a central minimum with a well depth confining particles with a kinetic energy below 0.5 kelvin equivalent [28]. A schematic for the superimposed Mixing and Atom Trap systems can be seen in Figure 3, including solenoids A, B and electrodes for the trapping and manipulation of constituent particles, as well as the mirror coils and octupole for recombination and trapping of the resulting anti-hydrogen.

2.3 The ALPHA-g Experiment

The ALPHA-g apparatus is located adjacent to the Mixing and Atom Trap. Positrons and anti-protons are injected in the apparatus, mixed and then trapped between two solenoids and an octupole, in a similar set up to ALPHA-2, but instead in a vertical configuration. The anti-hydrogen atoms are then confined to the minimum of the potential profile described by

$$U(\vec{r}) = -\vec{\mu} \cdot \mathbf{B} + m_{\bar{H}} \vec{g} \cdot \vec{r}, \quad (6)$$

where $\vec{g}$ and $\vec{r}$ are the gravitational acceleration and position vectors respectively, and $m_{\bar{H}}$

is the mass of anti-hydrogen. A cross-section schematic for this setup can be seen in Figure 4. The current in the two solenoids is then decreased slowly, allowing anti-hydrogen to escape through the lowest saddle-point in the confinement field which, in the assumption that anti-matter falls, would favor the lower end. If this 'ramping down' is done at different rates for each solenoid, the probability of escape from the top or bottom can be manipulated. For example, if the bottom solenoid was configured to be stronger by a potential

$$\Delta U = m_{\bar{H}} g_z \Delta z / \mu_z, \quad (7)$$

where Δz is the vertical separation of the two solenoids, and the subscript z indicates a component on the z -axis, then the probability of escape from the top or from the bottom would be equal. The resulting escape probabilities are measured for varying coil strengths [34], and the equilibrium point identified (through measuring annihilations on the trap walls at the top and bottom simultaneously) yielding a measurement for $m_{\bar{H}}$, or rather anti-hydrogen's gravitational mass; As well as to test the behavior of anti-hydrogen against simulations. Annihilations on the trap wall may be measured through the use of a time projection chamber, designed to reconstruct the paths of anti-hydrogen through the tracking of released charged pions. In this project, our aims are to explore how the behavior of particles may be simulated, and to apply these techniques to a system resembling the ALPHA-g apparatus.

3 Simulating Particle Motion

This section covers the various numerical techniques and probability distributions we have explored in this project, and our employment of these techniques in a program to simulate the escape of neutral particles undergoing stochastic heating from a potential well.

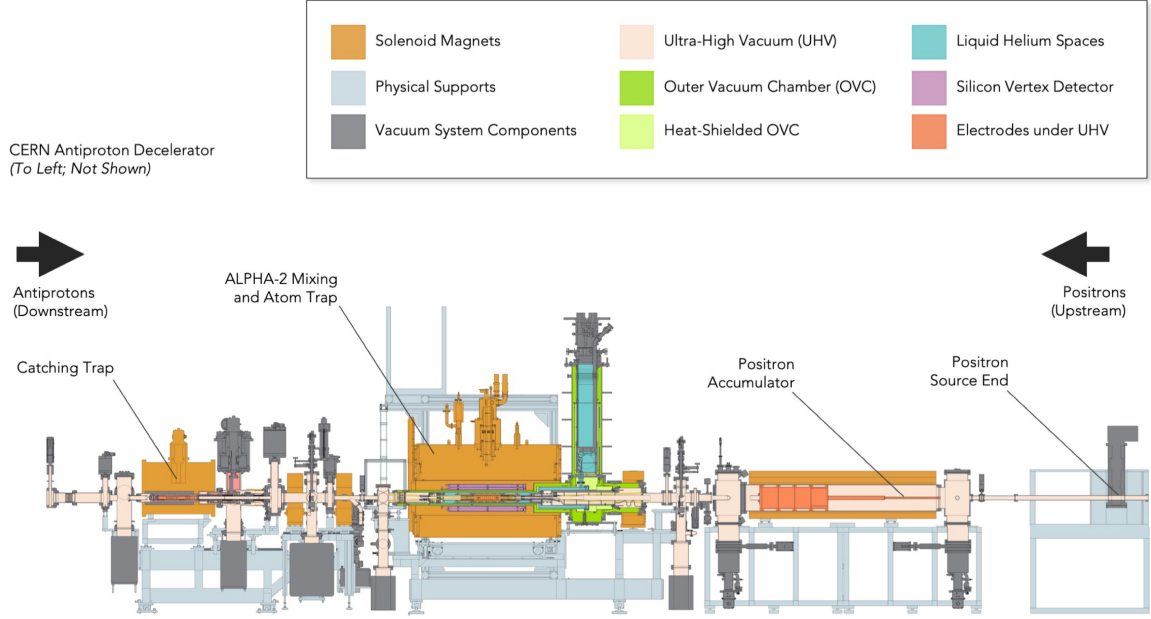


Figure 1: An overview schematic of the ALPHA-2 apparatus in its configuration from 2017 to 2019. Shown are the Catching (left), Mixing and Atom (middle) Traps as well as the Positron Accumulator and Source (right). The Catching Trap receives anti-protons from the Anti-Proton Decelerator off to the left. Figure provided by [28].

3.1 Randomly Distributed Variables

An important technique in particle simulations is the determination of quantities that are characterized by some distribution (for example, the Maxwell-Boltzman distribution), and are important for producing realistic predictions of future evolutions from real world data. In our project, we do this through a sampling of the distribution's cumulative probability distribution function (CDF), as the CDFs and probability density functions (PDFs) for all random variables we use are well known. To generate N randomly distributed variables from a CDF, we first generate N uniform random numbers between 0 and 1 inclusively. These random numbers are then interpreted as the cumulative probability, and the CDF is solved at this value for the randomly distributed variable; resulting in a distribution in line with the variable's PDF. An example of this process is the random distribution of velocities subject to the Maxwell-Boltzman dis-

tribution, where the CDF and PDF are,

$$\text{CDF}(\epsilon, T) = \text{erf}\left(\sqrt{\frac{\epsilon}{k_B T}}\right) - \frac{2}{\sqrt{\pi}} \sqrt{\frac{\epsilon}{k_B T}} \exp\left(-\frac{\epsilon}{k_B T}\right), \quad (8)$$

$$\text{PDF}(\epsilon, T) = \sqrt{\frac{2}{\pi}} \frac{2\epsilon\sqrt{m}}{(k_B T)^{3/2}} \exp\left(-\frac{\epsilon}{k_B T}\right), \quad (9)$$

where ϵ, m are the total energy and mass of the particle, k_B is the Boltzmann constant, T the temperature of the distribution and all other symbols or functions have their usual meanings [36]. The result of this process can be seen in Figure 5, showing this method produces a distribution of velocities that matches the PDF of the Maxwell-Boltzmann distribution. Notably, the CDF of the Maxwell-Boltzmann distribution is not solvable directly, as must be solved using a numerical method, as discussed in section 3.2. Another example is pulling randomly distributed 3D unit vectors, where the angles against the pole-axis (denoted here as θ) must be distributed to have a higher probability density around the equator in order to avoid

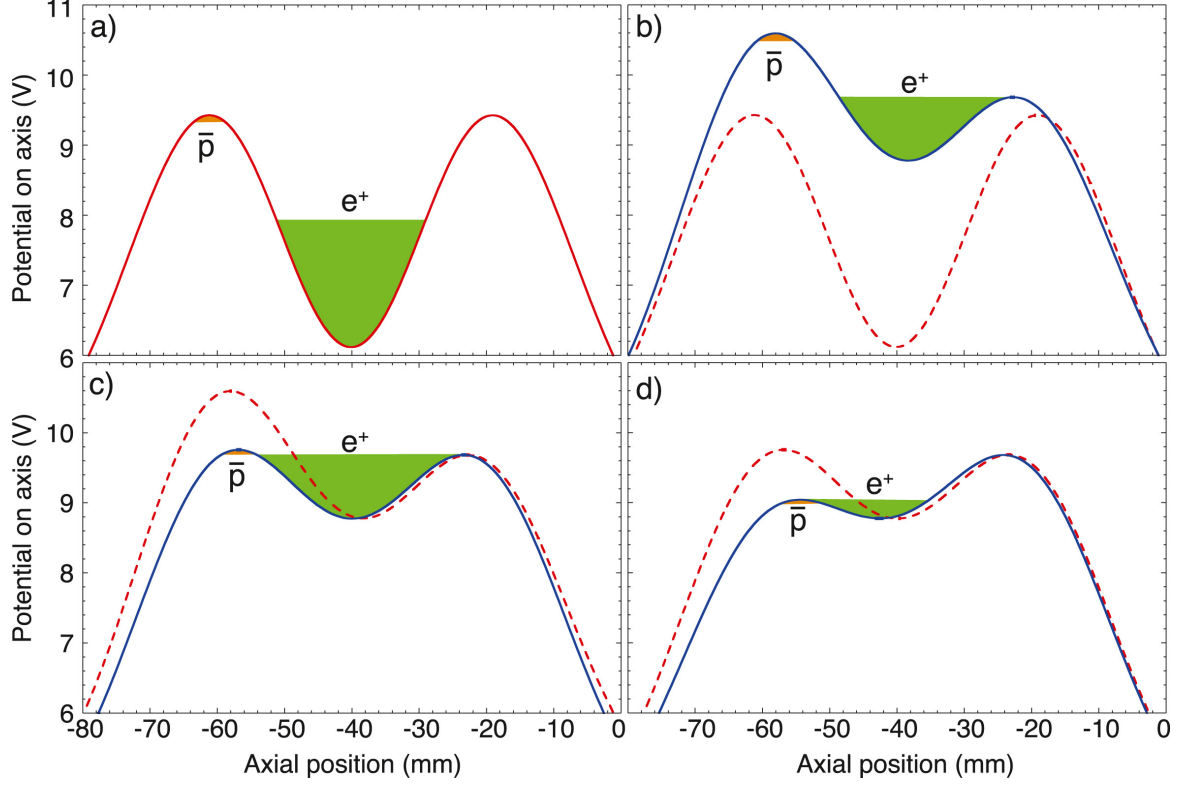


Figure 2: On axis Penning-Malmberg trap electrode potentials at different steps in the mixing process. The solid line represents the potential along the axis of the trap. The dashed line represents the potential profile in each previous step for comparison. Profile a) describes the potentials pre-mixing, b) shows the evaporative cooling the positrons, during which some escape to the right. c) and d) then show the two plasmas in preparation for mixing and during mixing respectively, where the potential profile is manipulated to allow the positrons and anti-protons to interact; during which positrons can escape to the left, causing further evaporative cooling. Figure provided by [28].

clumping of vectors around the poles. In polar coordinates, the angles ϕ (azimuth around the z -axis) and θ must be distributed as,

$$\begin{aligned}\phi &= U[0, 2\pi), \\ \theta &= \cos^{-1}(1 - 2U[0, 1)),\end{aligned}\quad (10)$$

where $U[a, b)$ denotes a uniform randomly distributed variable between a and b inclusively. This results in θ angles with a higher tendency around the equator, as opposed to uniformly distributed angles ($\theta = U[0, \pi)$). The difference in the two distributions can be seen in Figure 6.

3.2 Numerical Methods

In numerical analysis, leapfrog integration is a second order numerical method for integrating

differential equations; where velocities and positions are updated at different time intervals in such a way that they "leapfrog" over one another. It is a commonly used method for time evolving dynamical Monte-Carlo simulations due to its reversibility and that it conserves the Hamiltonian of a dynamical system, or that it is symplectic in nature [37]. The equations for obtaining integer steps of position and velocity are,

$$\begin{aligned}\vec{a}_i &= \vec{F}_i(\vec{x}_i, t)/m, \\ \vec{v}_{i+1/2} &= \vec{v}_{i-1/2} + \vec{a}_i \Delta t, \\ \vec{x}_{i+1} &= \vec{x}_i + \vec{v}_{i+1/2} \Delta t,\end{aligned}\quad (11)$$

where Δt indicates a constant time step between two integer iterations, i indicates the iterative position, and $\vec{F}(x, t)$ is a (potentially time evolving) force acting on the

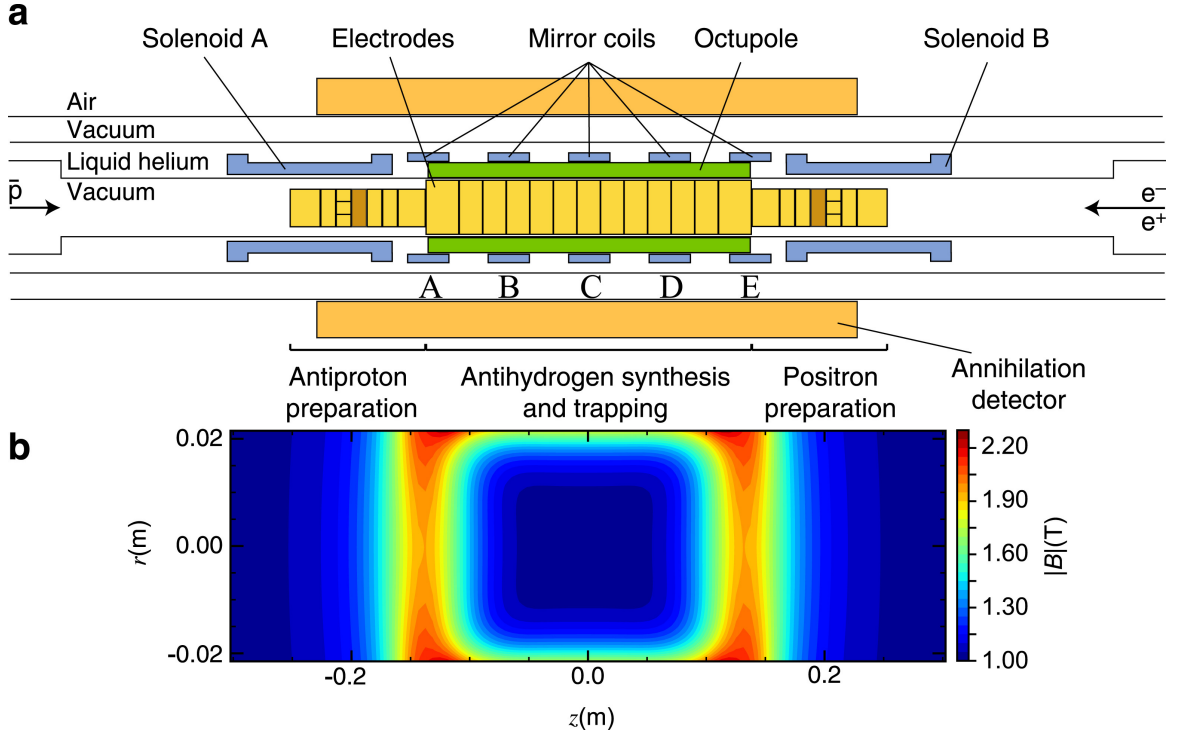


Figure 3: A schematic diagram for the Mixing and Atom Trap systems respectively. Shown in (a) are solenoids A and B, and the stacks of electrodes used for the preparation and manipulation of constituent anti-particles in the Mixing Trap, as well as the mirror coils and octupole coil used to manipulate and confine anti-hydrogen in the Atom Trap (where solenoids A and B are also used). Shown in (b) is magnetic field cross-section profile within the Atom Trap, showing a minimum in the centre where anti-hydrogen is confined. Note the presence of a background field of $1T$ across the Mixing and Atom Trap. Figure provided by [28].

system giving the characteristic acceleration; all other symbols have their usual meanings. These equations are stable for a time-evolving force as long as the rate of evolution is small over the duration of Δt .

Another numerical method common to simulations is iterative root solving for equations that are not analytically solvable; for example, the Maxwell-Boltzmann CDF (as discussed in Section 3.1). For our project we chose to utilize the Newton-Raphson (NR) method whenever we encountered this issue; the NR method follows the equations,

$$\begin{aligned} x_1 &= x_0 - f(x_0)/f'(x_0), \\ x_{i+1} &= x_i - f(x_i)/f'(x_i), \end{aligned} \quad (12)$$

where $f(x_n)$ and $f'(x_n)$ are the function to be solved and its derivative respectively evaluated at x_n , the subscript n denotes iteration and x_0 is an initial guess [37]. We chose this method due to it being quick to resolve a root to a high

degree of accuracy, as well as it requiring the derivative of the subject function; as the PDF and CDF of all distributions we are concerned with are all well-defined. One limit of concern is when the function or its derivative approach zero, leading to slow root finding or a divergence respectively; however, this only occurs for all our distributions at values equal to 0 or 1, which is not a statistically significant occurrence.

3.3 Stochastic Heating Simulation

In this project we aim to simulate the escape times of anti-hydrogen atoms trapped between two solenoids while undergoing a stochastic heating process. We aim to simulate non-adiabatic heating, where there is direct heat flow into our dynamical system as opposed to compressive heating; adiabaticity can be broken through rapid transitions in

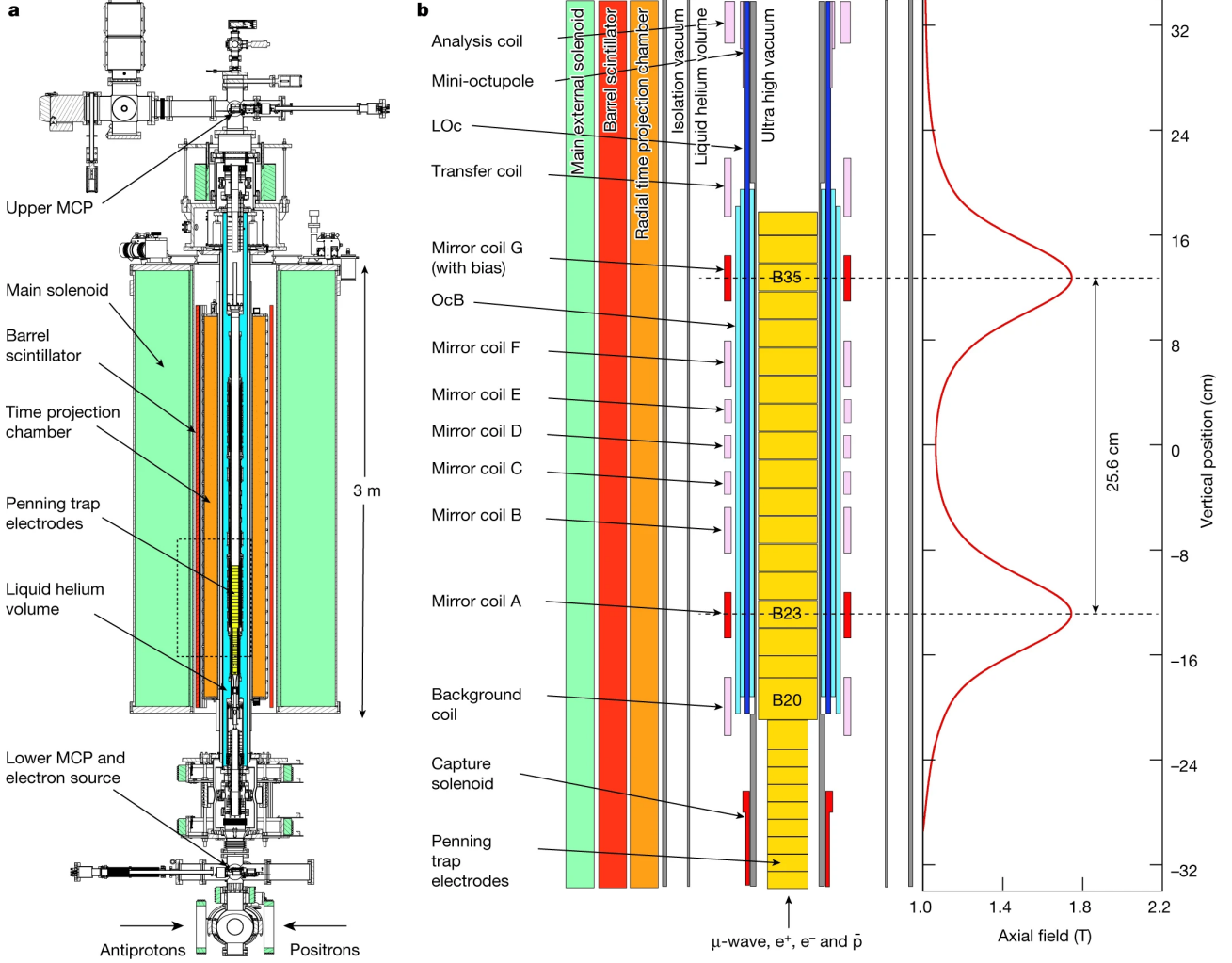


Figure 4: A cross-section schematic of the ALPHA-g apparatus (a) and a close up schematic along with the magnetic potential profile (b) provided by [35]. Shown is the injection of constituent particles, along with an electron source for electron cooling, and the main body of the trap inside a bath of liquid helium for cooling. The main body of the trap (the dotted line section in a) is shown in b, and consists of the Penning-Malmberg trap electrodes, solenoids and mirror coils as discussed in Section 2.2, as well as a time projection chamber. Also shown on the right is the axial magnetic field profile as a function of vertical position in the trap.

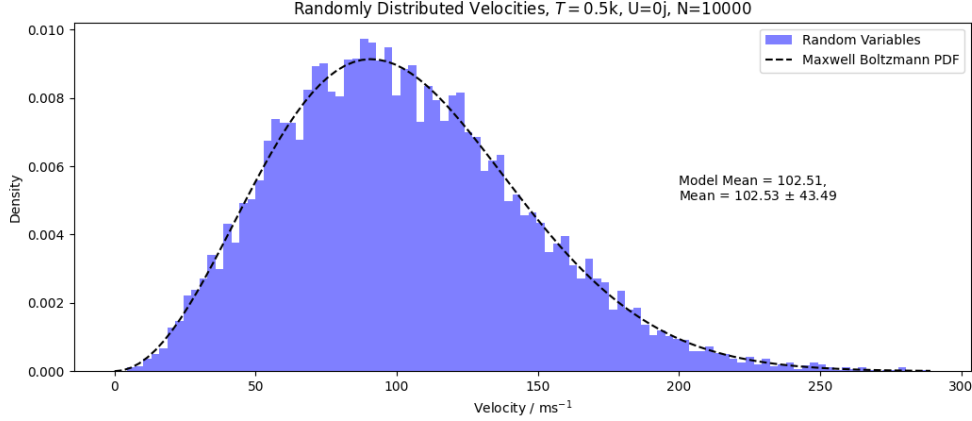


Figure 5: A normalized histogram of 10000 randomly distributed velocities for a Maxwell-Boltzmann distribution at a temperature of 0.5 Kelvin, for a particle mass equal to that of a hydrogen atom, as discussed in Section 3.1. The histogram of particle velocities is shown in blue and the distribution's PDF is overlaid with a black dotted line. The mean velocity of the Maxwell-Boltzmann distribution at this temperature is 102.51 ms^{-1} , and the mean and standard deviation of the distributed variables is $102.53 \pm 43.5 \text{ ms}^{-1}$; this is well within one standard deviation of the model.

potential, leading to changes in the energy in the system. Our simulation consists of a cylindrical trap, of length and diameter L and d respectively, enveloped in a radially centralized magnetic potential and contained within an axial potential well, similar to the solenoid/octupole set-up of ALPHA-g. The particles are modelled to be at some initial temperature T_i , leading to velocities distributed as per a Maxwell-Boltzmann distribution (described in Section 3.1).

Spanning the length of the trap on axis is a Gaussian standing electromagnetic beam [38], of wavelength λ , beam-power P and beam width σ . A Gaussian beam has the electric field equation,

$$E(\vec{r}, z, t) = E_0 \frac{\sigma_0}{\sigma(z)} \exp \left[-\frac{r^2}{\sigma^2(z)} \right] \times \exp \left[i(kz - \psi(z) + \frac{kr^2}{2R(z)}) \right] \times \exp \left[-i\omega t \right], \quad (13)$$

where r is the radial distance from the center of the beam, z is the axial distance from the beams "focus" (point of minimum beam width), k is the wave number, σ_0 is the minimum beam width, E_0 is the electric field am-

plitude, $\sigma(z)$ is the radial beam width at which the amplitude falls to a factor $1/e$ at z , $R(z)$ is the curvature of the beam's wavefronts at z and $\psi(z)$ is the Gouy phase, attributable to the phase velocity of light. For the simplicity of our simulation, we combine the wavefront and Gouy term into a combined phase term $\exp(i\phi(z))$. The beam width $\sigma(z)$ is defined by the equation,

$$\sigma(z) = \sigma_0 \sqrt{1 + (z/z_R)^2}, \quad (14)$$

where z_R is the Rayleigh range of the beam. In our simulation, we also approximate that the anti-hydrogen atoms are confined to a region much smaller than the Rayleigh range, and that the radial amplitude of the beam is a constant inside the beam width and zero outside, such that $\sigma(z) = \sigma_0$ (henceforth σ), and

$$\frac{\sigma_0}{\sigma(z)} \exp \left[-\frac{r^2}{\sigma^2(z)} \right] = 1 \text{ for } r < \sigma, \quad (15) \\ = 0 \text{ for } r > \sigma.$$

The superimposed standing wave resulting from two otherwise identical waves travelling in the opposite direct with opposite phase ϕ is then,

$$E(z, t) \propto \exp(-i\omega t) \cos(kz). \quad (16)$$

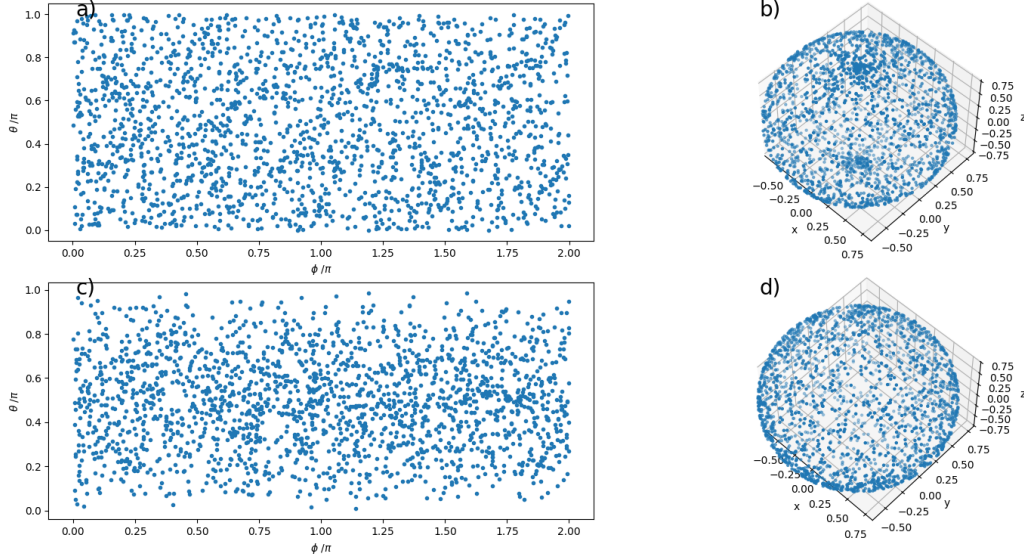


Figure 6: A graphical representation of the differences between a uniform θ (angle against the z -axis) in a, b and a correctly distributed θ angle in c, d for 2000 randomly distributed θ, ϕ . Plots a and c show a 2D projection of how θ changes with azimuth angle ϕ , with those angles then projected onto a 3D unit sphere in plots b and d. Note that for a and b, the uniform density of angles across θ leads to a higher clumping of angles around the poles of the unit sphere, where the correctly distributed angles in c and d lead to a uniform surface density in random unit vectors.

The force experienced by an atom in an electromagnetic field is,

$$F = \frac{1}{2}\alpha\nabla E^2, \quad (17)$$

where α is the polarizability of the atom [39], and the force is experienced along the direction of propagation of the beam. We use the relationship between electromagnetic amplitude and beam power to give us the relation,

$$\delta U = \frac{\alpha}{4\pi\epsilon_0} \frac{4P}{c\sigma^2} [1 - \cos^2(k\Delta z)], \quad (18)$$

to describe the change energy of the atom as it crosses and axial length Δz , where ϵ_0 is the polarizability of free space, and c is the speed of light in a vacuum.

This equation shows that a particle would not experience any change in energy as it traverses a whole wavelength of the beam. To subvert this relation and break the adiabaticity of the system, we simulate rapidly and instantaneously randomly changing the phase of the laser at some constant time interval τ_{phase} ; resulting in the change in energy of the particle

during this time interval to be permanent. We call this change in energy every time interval τ_{phase} a 'kick' in energy. The phase of the beam is modelled as a random uniform variable, so we take the average of the cosine term to find the average energy deposited onto the particle for each kick,

$$\delta U = \frac{1}{2} \frac{\alpha}{4\pi\epsilon_0} \frac{4P}{c\sigma^2}. \quad (19)$$

As the particle traverses the beam over a time interval τ_{beam} (equal to $2\sigma/v_t$, where v_t is the particle's transverse velocity), it will experience $\tau_{\text{beam}}/\tau_{\text{phase}}$ total kicks of energy; these kicks can be either a positive or negative change in energy with equal probability, hence we model this as random walk in energy for a particle's traversal of the beam.

An integer random walk describes a random process where a variable is incremented by ± 1 with equal probability, such that,

$$P_{i,i+1} = P_{i,i-1} = 1/2, \quad (20)$$

where $P_{a,b}$ denotes the probability of a transition from state a to b . The probability of

i equalling any number can be found using a normalized Pascal's triangle; where for N total steps the probability that a random variable S_N is,

$$P(S_N = n) = 2^{-N} \binom{N}{(N+n)/2}, \quad (21)$$

where the bracketed term denotes the number of ways of choosing $(n+k)/2$ elements from a set of n total elements [40], and is written from a factorial,

$$\binom{N}{(N+n)/2} = \frac{N!}{((N+n)/2)!(N - (N+n)/2)!}. \quad (22)$$

The central limit theorem motivates that this distribution tends to a normal distribution in a large N limit, becoming,

$$P(n, N) = \frac{1}{\sqrt{2\pi N}} \exp\left(-\frac{n^2}{2N}\right), \quad (23)$$

with a mean 0 and a standard deviation $\sqrt{N}$; a demonstration of this limit can be seen in Figure 7 for $N = 10000$. This model describes the probability of receiving n non cancelled 'kicks' while traversing the beam for N total 'kicks', and gives the total energy change in the particle. After receiving a change in energy and exiting the beam, we model the particle redistributing its new velocity over its degrees of freedom, in order to simulate the effect of the potential profile inside the trap coupling its components of motion.

Inside the trap, the particle observes the beam occupying a cross-section equal to $2\sigma L$, from this we can use the mean free path of a particle to model the probability of it crossing the beam after some time t ; the probability of a particle having crossed the beam after some time t is described as,

$$P(t) = 1 - \exp\left(-\frac{v_t t}{\lambda}\right), \text{ where } \lambda = \frac{\pi d^2}{2\sigma}, \quad (24)$$

where λ is the mean free path of the particle and v_t is its cross-sectional velocity. We assume that the particle will only redistribute its velocity once in the timescale between beam passes, and hence that v_t is a constant until

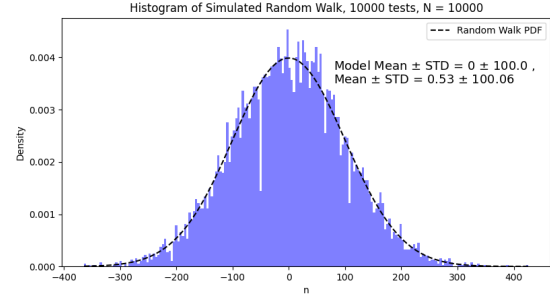


Figure 7: A histogram showing a simulated random walk for 10000 variables over 10000 steps. The mean of the simulated variables is 0.53, with a standard deviation 100.06; this shows that the normal distribution is indeed a sufficient description for a large N random walk.

the particle next crosses the beam. From this theory we can run a basic simulation for an ensemble of particles crossing a beam at random intervals and being randomly heated until they gain energy over a threshold, to model escape times for anti-hydrogen atoms stochastically heated while trapped inside a potential well.

Another variable effecting the number of kicks our particle may receive is its cross-sectional impact parameter b ; this is found for a particle a distance r from the beam, with an angular path θ against the vector of the shortest approach to be,

$$b = r \sin(\theta). \quad (25)$$

Since the trap diameter is much larger than the beam diameter, we can assume that r is large and that hence θ is small, allowing us to utilize the small angle approximation for $\tan$ to pull a linear relationship between b and θ . The transverse angle of a particle is uniformly distributed, meaning the impact parameter is also a uniformly distributed random variable between 0 and σ for a large trap size.

4 Implementation and Results

Having derived the distributions and methods for our simulation, we chose to employ our simulation in Python for simplicity and proof of

concept, though for a more rigorous simulation I would recommend a more computationally efficient language such as C++. We employ a two-dimensional (non-spacial) Monte-Carlo simulation, where particles are tracked by their velocity magnitude and their time spent in the trap. We choose to iterate over the number of passes each particle has made of the beam, instead of the more conventional use of a time step; this is due to the time spent between beam passes being much longer than any step with a sufficiently small resolution in time for meaningful results, meaning for each time step the vast majority of particles would not be considered important and waste computation. A table of dimensions and properties of our simulated trap, beam and initial conditions is found in Table 1, these are arbitrary for proof-of-concept but are not nonphysical to typical real values [28]. The threshold energy was chosen to be a factor of ten larger than the initial distribution temperature to ensure the probability of a Maxwell-Boltzmann distribution at the initial temperature having a particle above the threshold temperature is statistically insignificant. The energy range of $1 - 10$ mK is equivalent to a particle velocity of the order $1 - 10$ ms $^{-1}$; hence the maximal number of kicks a particle will receive as it crosses the beam is of the order,

$$\frac{\tau_{\text{beam}}}{\tau_{\text{phase}}} = \frac{2\sigma}{v_t} \frac{1}{\tau_{\text{phase}}} \approx 10^4, \quad (26)$$

allowing us to assume a large N limit for random walk statistics.

The simulation loop is designed to run as follows; X particles are initiated at T_i with velocities following a Maxwell-Boltzmann distribution as $\mathbf{V}$, each with an attributed total number of beam passes, local time and total accumulated number of kicks n_{tot} all initiated at 0 as tracking variables. Here the bold symbol indicates a vector of X values coinciding with X particles. Then X random unit vectors are generated, with transverse and axial components calculated as $\hat{\mathbf{T}}$ and $\hat{\mathbf{z}}$ respectively, allowing us to find the transverse components of all particles as $\mathbf{V}_t = \hat{\mathbf{T}} \cdot \mathbf{V}$. A randomly distributed time until the next beam pass is then

pulled for each particle, using the mean free path probability distribution and the method described in Section 3.1, and each local time iterated by its respective change. X impact parameters are then uniformly distributed, and used to calculate the total number of kicks each particle will receive as a function of impact parameter and its axial velocity,

$$N = \frac{2\sqrt{\sigma^2 - b^2}}{\mathbf{V} \cdot \hat{\mathbf{T}}} \frac{1}{\tau_{\text{phase}}}. \quad (27)$$

Numbers of accumulated kicks $\mathbf{n}$ is then randomly distributed using the distribution above, using the same method described previously, and from that a positive or negative energy change for each particle in this beam pass is calculated as,

$$\Delta U = \mathbf{n} \delta U. \quad (28)$$

A difference in velocity is then calculated for each particle and is added onto its z component, and a new total velocity calculated. At this point the simulation checks if the energy of each particle has exceeded the given threshold, and if so, records its current local time and kills the particle. After a minimum number of escapes has occurred, the simulation checks if any particle has exceeded a maximum time constant t_{max} at which point the particle is assumed to be 'stuck', or at least not able to escape in any reasonable time frame, and is also killed. Tracking variables are then iterated, and this loop repeats until a maximum number of iterations/beam passes is reached (at which point any remaining particles are also assumed to be stuck), or all particles escape.

We ran this simulation for groups of 20,000 particles, recording the velocity and accumulated kick distributions, as well as escaped particles, at $t = 0, 1000, \dots, 5000$ s, before letting each particle run until $t = 10000$ s (about 3 hours) or the whole sim until 10000 beam passes where remaining particles timed-out, whichever happens first. A histogram showing the velocity distribution of the particles at $t = 0$ s through 5000 s can be seen in Figure 8. This histogram shows clearly the mean of the velocity distribution increasing with time, from

$\bar{v} = 4.58\text{ms}^{-1}$ at $t = 0$ s to $\bar{v} = 9.32\text{ms}^{-1}$ at $t = 5000$ s, showing indeed that on average, particles are being heated from a low temperature by our model. The standard deviation over time first increases from the initial, but then begins to decrease beyond $t = 2000$ s; the initial increase is due to the nature of the Maxwell-Boltzmann distribution, whereas the mean temperature of the distribution increases it begins to flatten out. However, as the mean temps upwards, the higher energy tail of the distribution begins to get cut off by the energy threshold line, leading to an overall decrease in standard deviation as the simulation tends on. A plot of the mean distribution velocity over time can be seen in Figure 9; these data points have been fitted to an exponential model using the least squares fit as,

$$\bar{v} = b \exp \left[\frac{t}{a} \right] + d, \quad (29)$$

where $a = 3112 \pm 187$ s, $b = -5.91 \pm 0.17 \text{ ms}^{-1}$, $d = 10.51 \pm 0.18\text{ms}^{-1}$. The fit of this model shows that the rate at which the distribution mean increases is decreasing, and will tend to an asymptotic limit at $\bar{v} = 10.51 \pm 0.18 \text{ ms}^{-1}$. This can be attributed to the relationship between an increase in energy and a change in velocity being inversely proportional to velocity,

$$E = \frac{1}{2}mv^2 \rightarrow \frac{dv}{dE} = (mv)^{-1}, \quad (30)$$

and that each particle will spend less time in the beam as its velocity increases, leading to lower numbers of kicks. It is worth noting that this limit is below the threshold energy of the simulation (1 mk is equivalent to $\approx 13 \text{ ms}^{-1}$), however particles will still continue to escape as they are distributed around this mean, until the probability of a particle having a velocity over the threshold becomes statistically improbable for the number of particles remaining in the system.

A normalized histogram of escape times can be observed in Figure 10; this data has been fitted to a model described by,

$$y = b \exp \left[\frac{t}{a} \right], \quad (31)$$

Property	Symbol	Value
Trap Diameter	d	5×10^{-2} m
Beam Diameter	σ	200×10^{-6} m
Beam Wavelength	λ	1200×10^{-9} m
Beam Power	P	2 W
Beam Switch Time	τ_{phase}	1 ns
Initial Dist Temp	T_i	1 mK
Threshold Energy	T_f	10 mK

Table 1: A table showing constant properties used inside our simulation. Note the value of Threshold Energy being given in temperature equivalent units ($E = k_B T$ where k_B is the Boltzmann constant).

where $a = -7.74 \pm 0.2 \times 10^3$ s, $b = 1.78 \pm 0.02 \times 10^{-4} \text{ s}^{-1}$. This model describes a clear but small (with respect to timescale) decrease in the rate of particle escape as time increases. This implies that the rate of escape of particles is inversely proportional to the number of remaining particles, which is a behavior we would expect for particles who are subject to a probability distribution. From this model we can define a "mean escape time" $\bar{\tau}$ which would be equal to the absolute value at a , 7740 ± 200 s, the time when number of particle still in the trap has dropped by a factor $1/e$.

At the end of the simulation, after either 10000 s or 10000 passes of the beam, 3079 particles remained stuck in the simulation. The distribution of effective kicks for these particles, and for the escaped particles, is shown in Figure 11. The standard deviation of kicks for the stuck particles is higher than for the escaped, implying that on average, escaped particles actually received fewer effective kicks than those stuck at the end of the simulation. From this, we can infer that the stuck particles are primarily those who were 'unlucky' in the initial distribution of velocities, and needed to accumulate more kicks in order to escape the simulation; whereas particles who were distributed with a higher velocity could escape more readily.

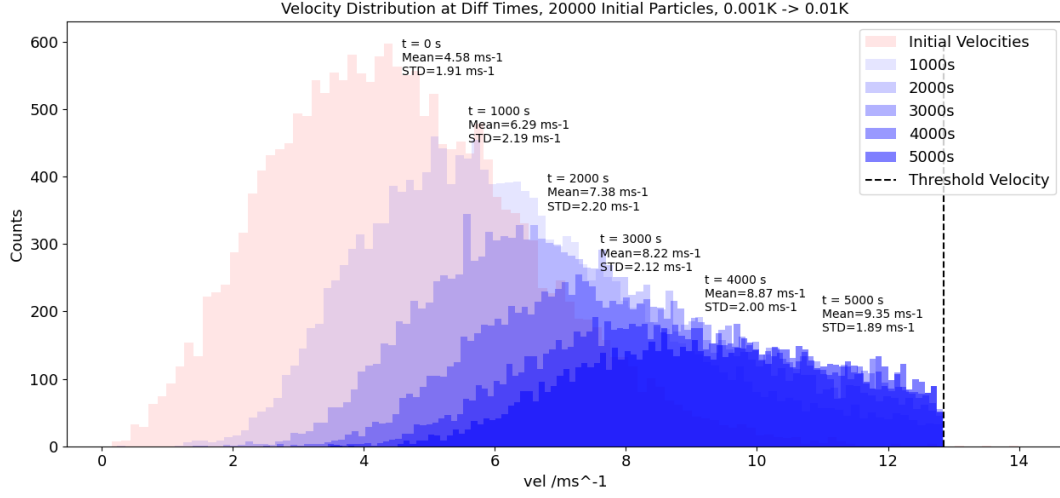


Figure 8: A histogram showing the distribution of particle velocities at $t = 0, 1000, \dots, 5000$ s, with the initial velocities in salmon, and the velocities at every proceeding time plotted progressively bluer with higher opacity. A text label displaying the time the velocities were taken, as well as the mean and standard deviation of each distribution is displayed adjacent to its corresponding distribution. A dotted line representing the threshold energy of 0.01 K temperature equivalent is at the far right of the graph. Notice the lack of any particles further right than this line, as these have all 'escaped'. Notice also how the total area covered by each distribution decreases as time goes on, due to more and more particles escaping over time; with 17096, 14684, 12588, 10675, 8946 particles remaining at each time from $t = 1000$ s respectively.

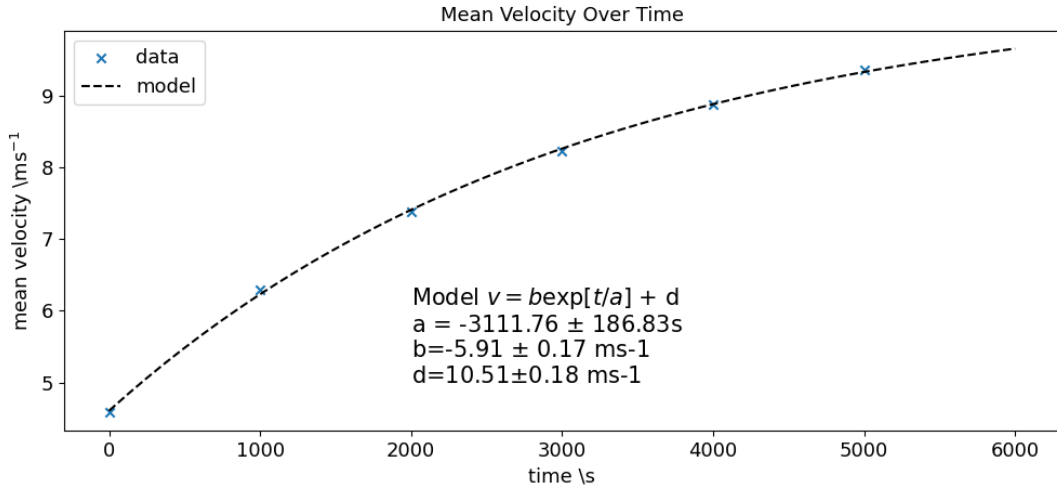


Figure 9: A plot of the mean velocity over time from results presented in Figure 8. The data point have been fitted to an exponential model as in equation 29; showing the change in velocity over time to be decreasing to a asymptotic limit at $10.51 \pm 0.18 \text{ ms}^{-1}$.

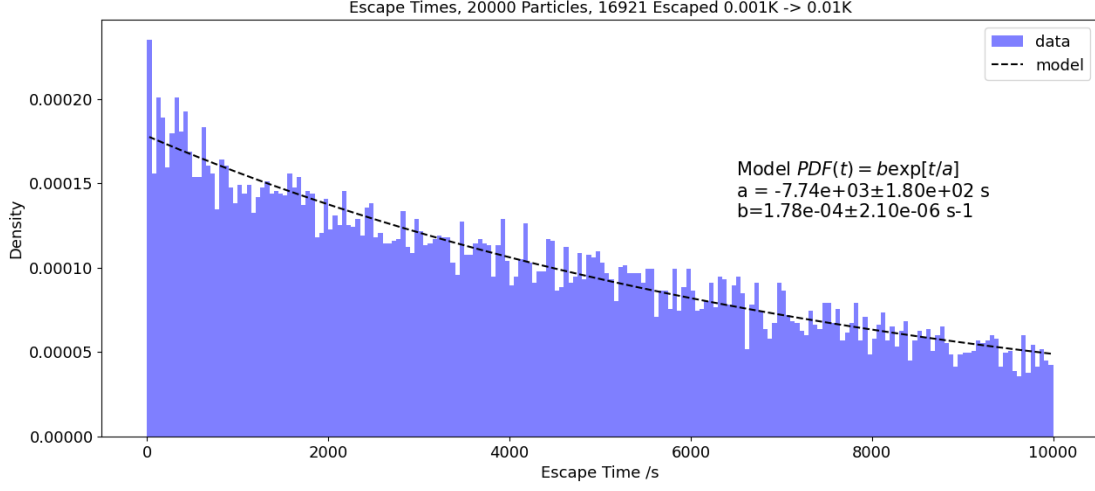


Figure 10: A histogram of counts for times at which particles escaped the simulation (blue). Over-layed in the dotted black line is an exponential model fitted to the available data, as described by Equation 31.

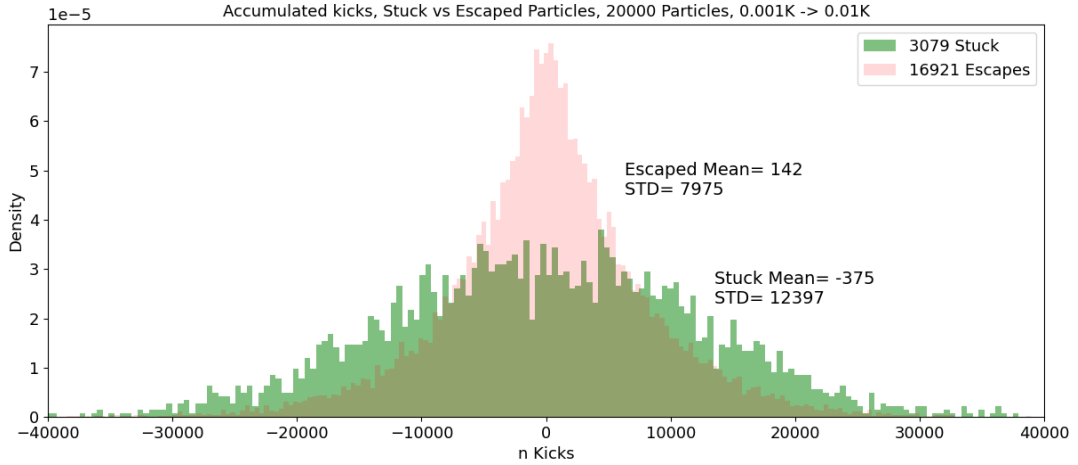


Figure 11: Normalized histograms showing the distribution of accumulated kicks for the 16921 escaped particles (pink) and the 3079 stuck particles (green). The mean and standard deviation for the escaped particles are 142 and 7975 respectively, whereas the mean and standard deviation for the stuck particles are -375 and 12397 respectively.

5 Future Work

There are a number of ways this project can be further developed but, due to the constraints of time, our ambitions had to be somewhat reduced. Most ideally, we would've like to have repeated the simulation a multitude of times, for a greater number of particles, allowed for a full statistical treatment for optimizing our models describing the change in mean velocity of the distribution and escape times. Paired with this, a full analytical treatment would give us theoretical values for quantities such as the asymptotic temperature limit and the "mean escape time" to compare our simulation to. Additionally, we would've liked to observe the behavior of the distribution for different rates of heating (perhaps by varying P or τ_{phase} over a range), which we believe would further tie in to the aforementioned analytical treatment, effecting the rate at which particles would heat up and leave the simulation. A further development would be to change the method of heating; for example, to mimic the effect of a gravitational potential on the trap, as would be the case in the ALPHA-g experiment. Such a change would be paired with a full spacial Monte-Carlo treatment, including the full magnetic potential profile of Penning-Malmberg trap; this would allow us to track the particle's position in the trap, and record the specific side a particle may escape or if it simply annihilates on the trap walls. Finally, we would then like to compare the result of such a simulation to the results from the real-world ALPHA-g experiment, allowing us to calculate the gravitational mass of anti-hydrogen and determine if gravity treats hydrogen and anti-hydrogen alike.

6 Conclusion

We have explored various methods and techniques for simulating the motion of particles and the transfer of energy in a dynamical system; these include numerical integration methods and randomly distributing variables subject to a distribution. We have applied these methods to simulate the non-adiabatic heating of an ensemble of anti-hydrogen-like atoms

trapped inside a potential well, with a distributed initial velocity. Furthermore, we observed the escape times for these particles and tracked the distribution of their velocities as the simulation progressed, and applied crude models to the results, in order to speculate on the behavior of an analogous real-world system and to draw a parallel to a theoretical treatment of the system. We found that our simulation did indeed result in the heating of the ensemble of atoms, and the applied model implied a tendency to an asymptotic limit at $10.51 \pm 0.18 \text{ ms}^{-1}$. Additionally, we observed the simulated escape times for the particles in this system, inferring that their distribution follows an exponentially decreasing model, and defined a "mean escape time" (where the number of particles remaining in the trap has dropped to a factor $1/e$) at $\bar{\tau} = 7740 \pm 200 \text{ s}$. Finally, we then speculated on the future path of this simulation and how we would tie this work into the real-world ALPHA-g experiment. All project code is available on GitHub, at <https://github.com/ECD-git/Sim-Grav-Motion-AntiHydrogen>. No generative AI was used in this project.

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